

Coordination Behavior of a Confined Ionic Liquid in Carbon Nanotubes from Molecular Dynamics Simulations

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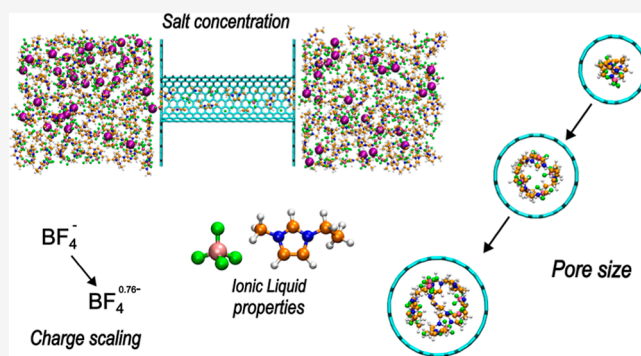


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ABSTRACT: To understand the behavior of ionic liquids (ILs) at carbon material, i.e., carbon nanotube (CNT)-containing pores, we simulated different systems and analyzed their structural—in particular their coordination—behavior as well as their velocity distribution. The extension of our analysis tool CONAN presented here allowed us to study the coordination behavior as a function of the distance to the carbon material. Our systems were composed of three different CNTs combined with either the neat IL 1-ethyl-3-methylimidazolium tetrafluoroborate or with their NaBF_4 salt mixtures. We investigated the impact of the force field charge scaling. As previously detected, the neat IL assumed radial layers within the confinement, with the radial density distribution depending strongly on the pore size. For the salt mixtures, the sodium cation remained in the bulk and was observed only once inside a tube. In all systems, the ions showed an overall decreased coordination behavior for regions in the bulk phase close to the carbon pore and within the confinement. The coordination number was always reduced with scaled charges. For charge scaling, higher dynamics was observed also in confinement. Interestingly, the average velocity of the atoms near the surface inside the confined space was higher than that in the center of the pore.



1. INTRODUCTION

Liquids at surfaces and in confined spaces behave significantly differently from their bulk phase, with the resulting physical properties being highly dependent on the type of liquid and solid.^{1–7} This is particularly important for energy devices where the liquid is in contact with an electrode. At the molecular level, these property changes result from liquid–electrode interactions and an imposed change in liquid–liquid interactions near the surface.^{5,6} Water,^{8–10} water-based electrolytes,^{11–13} ionic liquids (ILs),^{6,14–21} and other electrolytes^{22–24} are often studied in combination with solids such as carbon materials,^{2,9,15,16,22,23,25,26} transition-metal oxides,¹¹ metal–organic frameworks,^{20,27,28} or metal surfaces.^{12,29} ILs have several favorable properties as electrolyte materials, such as low volatility, a wide electrochemical window, and high thermal and chemical stability. However, the viscosities of most ILs are high at room temperature, accompanied by a slow diffusion of ions, and thus neat ILs usually have a relatively low ionic conductivity at ambient conditions.⁷ There are examples, where adding salt in high concentration leads to better ion transport despite the fact of a possible higher associated viscosity.^{30–33}

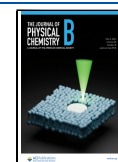
The study of the properties of such liquids at liquid–solid interfaces is of particular interest in the field of electrochemistry, with the aim of developing new types of energy storage systems and enhancing the desired properties by combining liquids and solids in a sophisticated way. For example, new electric double layer capacitors (EDLCs)^{25,34,35} with a higher energy density than conventional capacitors are sought, which are composed of carbon materials as electrodes in combination with ILs. Energy storage in these systems is based on the electrostatic attraction of charged particles, which allows rapid charging and discharging, but at the cost of a low energy density compared to conventional batteries.³⁶ The key to their energy storage capability is the formation of a Helmholtz double layer,³⁷ a concept central to describing how EDLCs work. When a voltage is applied, a double layer of charges forms at the interface between the electrode material

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and the electrolyte. This bilayer consists of two parts: one layer forms on the surface of the electrode, which carries an excess charge due to the applied voltage. The other one is the layer of ions adsorbed on the electrode surface (Helmholtz layer), which compensates for the excess charge. The Gouy–Chapman^{38,39} theory extends on this double layer concept. Instead of a distinct layer in the electrolyte, it proposes the existence of a diffuse layer that extends beyond a molecular layer and penetrates deeper into the bulk phase. The associated decrease in the potential is described as an exponential decay. Further developments to this effect were proposed by Stern⁴⁰ and later by Grahame,⁴¹ who proposed a combination of both theories. In the case of ILs and IL salt mixtures as electrolytes, there is an additional effect called overscreening.^{36,42–45} As the excess charge of the electrode is overcompensated by the first layer in the electrolyte, a second layer of counterions is formed behind the first layer. This resorting of the IL is not limited to two layers and can extend a few nm into the bulk at strong applied fields.³⁶

For the electrodes of EDLCs, porous carbon material is of particular interest because it provides a large surface area and thus a large interface with the liquid. The liquid also enters the porous carbon network, where the effect of the confined space on the properties of the liquid is highly dependent on the pore size in which it resides. It has been observed that the capacitance is greatly increased for systems where the pore size of the carbon electrode matches the size of the ions.^{18,26,46,47} Understanding the underlying effect of this phenomenon is key to the improvement of such EDLCs. However, the experimental observation of the molecular arrangement in such systems, especially within confinement, is challenging, and their computational modeling contributes to the knowledge of the governing effects at the molecular level.

The first step is to understand the behavior of ILs in a confined space without an applied field to the necessary depth to be able to draw precise conclusions for the next steps. New tools are helpful to analyze such anisotropic liquid–solid systems. CONAN,^{48–50} a program developed by our group that allows the generation of carbon structures, the setup of the simulation box, and the analysis of the simulation trajectory data, was created for this purpose. Besides the established analysis methods from our previous publication,⁴⁸ new features have been added to the code to analyze the coordination and velocity of ions in the liquid with respect to their distance from the solid structure.

In this work, we study the effect of confinement on the coordination behavior and dynamics of neat $[\text{C}_2\text{C}_1\text{Im}][\text{BF}_4]$ and its salt mixture with 25 mol % NaBF_4 in pores of three different sizes using classical molecular dynamics simulations at an atomistic resolution. In the case of the electrode material, a carbon pore is constructed by using a carbon nanotube (CNT) and adding a graphene sheet to each open end of the CNT. The pores were constructed using CONAN. The computational details together with the implementation of the coordination number (CN) and the velocity analyses are provided in Section 2 of this work. The results are presented and discussed in Section 3 with concluding remarks in Section 4.

2. METHODS

2.1. Computational Details. All classical MD simulations of the neat IL $[\text{C}_2\text{C}_1\text{Im}][\text{BF}_4]$ and the IL salt mixture (25% molar fraction of NaBF_4) were conducted with the LAMMPS

program package⁵¹ using the OPLS-AA force field.^{52,53} The specific atomic partial charges are listed in Table S2 in the Supporting Information. For the $[\text{C}_2\text{C}_1\text{Im}]^+$ -cation and the $[\text{BF}_4]^-$ -anion the refined CL&P parameters^{54,55} for ILs were used, while the parameters of Cole & Klein were utilized for the carbon pore.⁵⁶ Both the construction of the simulation boxes and the equilibration procedure are discussed in the Supporting Information. All simulations were carried out at a temperature of 350 K in a canonical ensemble, using a time step of 1 fs.

The Lennard-Jones potential and pairwise Coulombic interactions were computed up to a distance of 12 Å, with tail corrections applied to the Coulombic part for larger distances. A particle–particle–particle-mesh solver for long-range interactions up to an accuracy of 1.0×10^{-5} kcal mol^{−1} Å^{−1} was implemented.

Each system consists of a bulk phase containing 512 ion pairs and a carbon pore structure generated using CONAN.⁴⁸ The structures are pristine and consist of a CNT with a radius in the nanometer regime and carbon walls attached to both open ends. The studied pores were constructed with CNTs in an armchair configuration of three different sizes. The radius of the respective pores is defined by the number of carbon rings n building the circumference of (n, n) CNTs and is listed in Table 1 together with the pores radii (r_{CNT}), length (l_{CNT}), and the number of carbon atoms each pore is composed of.

Table 1. Features of the Investigated Pores^a

notation	(n, n)	$N_{\text{atoms}}^{\text{pore}}$	$N_{\text{atoms}}^{\text{CNT}}$	r_{CNT}	l_{CNT}
8	(8, 8)	1548	544	5.44	40.55
10	(10, 10)	1668	680	6.79	40.56
12	(12, 12)	1752	816	8.15	40.57

^a $N_{\text{atoms}}^{\text{pore}}$: total number of carbon atoms of each pore; (n, n) : used CNT; $N_{\text{atoms}}^{\text{CNT}}$: number of carbon atoms including the radius of the respective CNT: r_{CNT} and the length l_{CNT} , both given in Å.

The simulation box dimensions in x and y directions (x : 38.34 Å; y : 39.352 Å) are adjusted to fit the carbon sheets, modeling an infinite carbon surface through periodic boundary conditions. The carbon pore is kept rigid over the whole production run. With the modeled systems, detailed in Table 2, three CNTs of different size are investigated, together with either the neat IL (N) or IL salt mixture (S). For instance, N_{10}

Table 2. Notation and Composition of the Investigated Systems; mol %: Salt Mole Fraction; Pore: the Used Pore; l_{box}^z : Box Length in z -Direction of the Box

notation	mol %	pore	$l_{\text{box}}^z/\text{Å}$
N_8	0	8	137.266
N_{10}	0	10	135.955
N_{12}	0	12	135.041
S_8	25	8	121.820
S_{10}	25	10	120.475
S_{12}	25	12	119.957
N_8^{sc}	0	8	145.762
N_{10}^{sc}	0	10	144.452
N_{12}^{sc}	0	12	144.520
S_8^{sc}	25	8	130.257
S_{10}^{sc}	25	10	128.981
S_{12}^{sc}	25	12	128.919

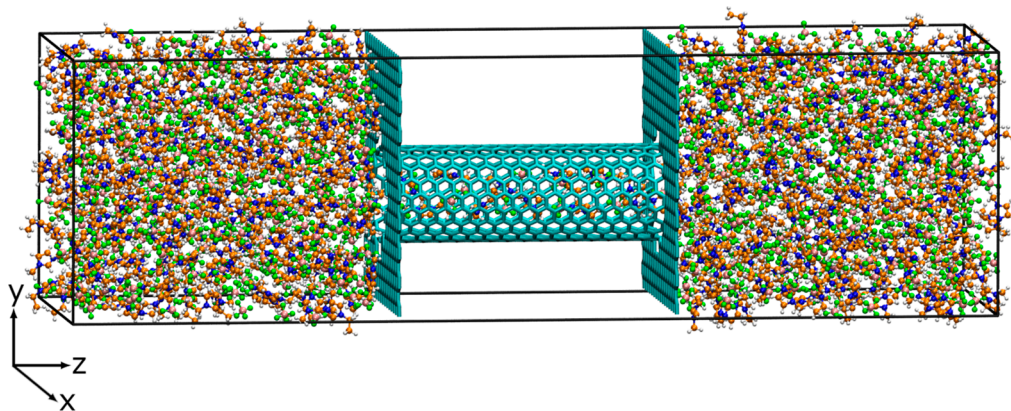


Figure 1. Ball-and-stick illustration of system N_{10} . Color code for the atoms; carbon: orange; nitrogen: blue; hydrogen: white; boron: pink; and fluorine: green. Carbon material: cyan and in a bond depiction; all was visualized using VMD.^{57,58}

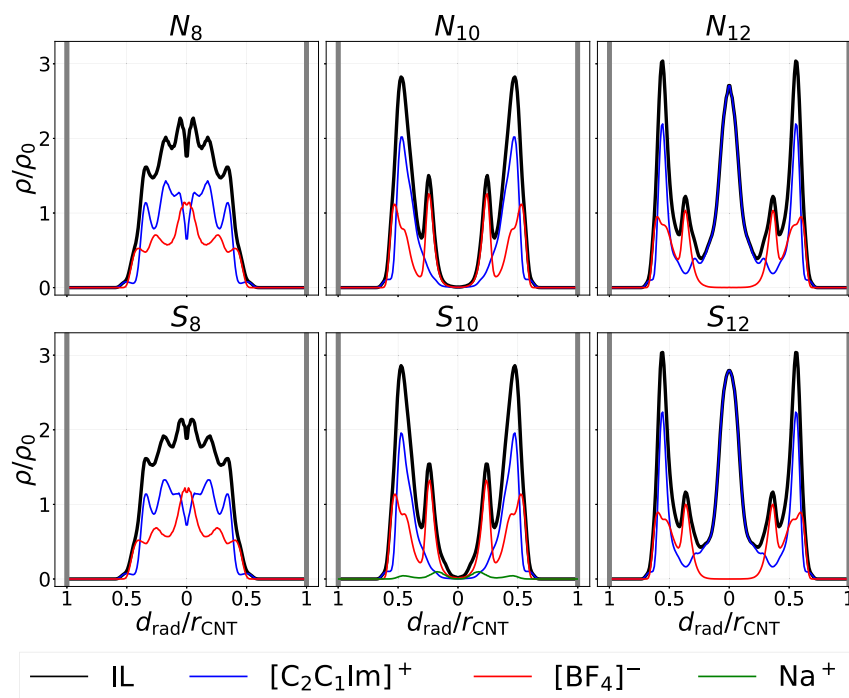


Figure 2. Radial mass density profiles for N_8 to N_{12} (upper panels) and S_8 to S_{12} (lower panels) of total and ionic contribution. Black: complete IL; blue: $[C_2C_1Im]^+$ -cation; red: $[BF_4]^-$ -anion; green: Na^+ -cation (only visible in case of S_{10}). The vertical lines represent the pore wall.

(Figure 1) is a system that contains neat IL and features pore 10. The mixtures, e.g., S_8 , were set up by exchanging 25% of the $[C_2C_1Im]^+$ -cations by Na^+ to obtain the desired 25% molar fraction of $NaBF_4$. This approach investigates the effect of the salt concentration on system properties.

In the first set of simulations, unit charges were used for the individual ions. In a second set, charges were scaled according to Stuchebrukhov and Leontyev.⁵⁹ The scaling factor is defined as the square root of the inverse of the dielectric constant ϵ_{el} . With the dielectric constant of the IL of 1.742,⁶⁰ the scaling factor of 0.76 was derived.

$$q_i^{sc} = \frac{q_i}{\sqrt{\epsilon_{el}}} = \frac{q_i}{\sqrt{1.742}} \approx 0.76 \cdot q_i \quad (1)$$

All systems with scaled charges are denoted by sc. During the first production run of the simulation (50 ns), every 1000th time step was saved. A second production run (10 ns) after the

first, saving every 250th time step, was run to compute the velocity distributions.

In order to explore the free energy landscape of a Na^+ -ion in our systems, we performed umbrella sampling for all of the pores. For this purpose, we set a radial spring potential that pulls a Na^+ -ion in 0.25 Å steps from an initial distance of 40 Å to the center of mass of the pore. The spring constant was set to 10 kcal mol⁻¹ Å⁻². After the production runs described previously, leaving all other modeling parameters unchanged, the Na^+ -ion was moved to the starting position over 0.25 ns. The start positions were determined by using the x and y coordinates of the end position (COM of a pore) and adding 40 Å to the z -coordinate. With the described approach of the radial spring potential in combination with the initial position, the energy profile depends mainly on the z -coordinate, but especially for the inside of the pore, also allows the cation to assume a radial position that is more favorable than on the central axis of the pore. For each increment studied, the Na^+ -

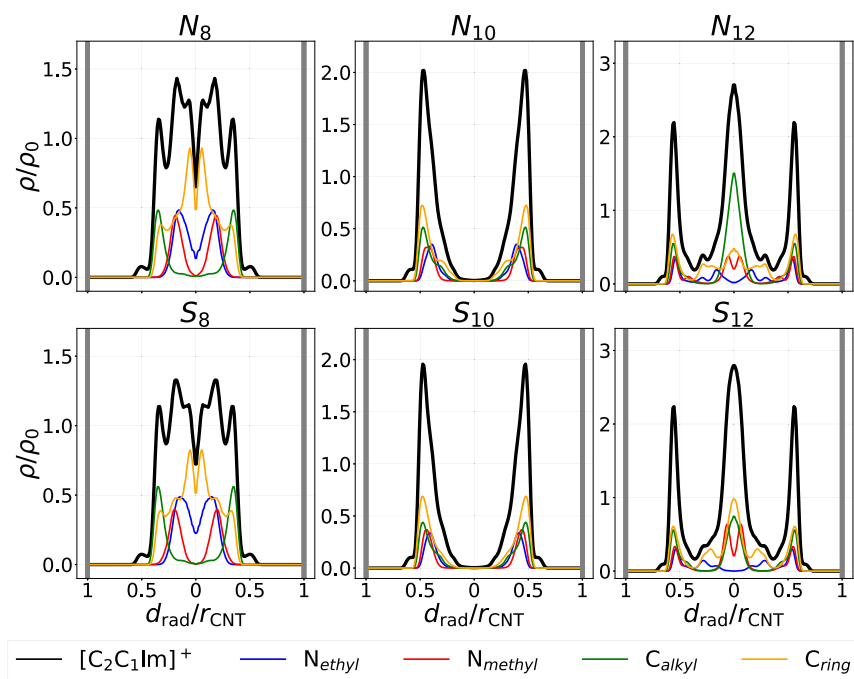


Figure 3. Radial mass density profiles for N_8 to N_{12} (upper panels) and S_8 to S_{12} (lower panels) of total $[C_2C_1Im]^+$ -cation and its atomic contributions. Blue: N_{ethyl} ; red: N_{methyl} ; green: carbon atoms of ethyl and methyl group; orange: carbon atoms of the imidazolium ring. The vertical lines represent the pore wall.

ion was positioned for a time of 0.1 ns before every 100th time step was stored for subsequent analysis for a further 0.1 ns. This results in 1000 data points for each of the 161 positions studied. The free energy curves generated from the collected data were obtained using the WHAM software.⁶¹

2.2. CN Analysis. To access the CN in confined space, a different approach from the number integral of the radial distribution function must be chosen because the immediate environment of a molecule depends on its position with respect to the porous carbon. Here, CN is obtained in the following way. First, a three-dimensional grid is adjusted to the simulation box with the incrementation chosen by the user. Next, the CN is computed by calculating the distances between the centers of mass of all molecules and counting how many particles are within a user-specified coordination distance. A distinction between the different types of molecules is made by default. The results are stored in Gaussian cube file format and can be visualized by appropriate visualization programs (VMD,^{57,58} PyMol⁶²). CONAN also provides the option to calculate the CN with respect to the distance to the nearest surface. This is done separately for molecules in the bulk and in the pore. The results are plotted as a heatmap. Averaging over the entire bulk or confined space is also an option.

To validate our approach, the CNs of the anions around the cations ($[C_2C_1Im]^+$) in systems N_8 – N_{12} and S_8 – S_{12} were calculated for a selected bulk part and compared to the number integral. The number integral was calculated with Travis.^{63,64} The results are shown in the Supporting Information in Figure S13. It can be seen that both approaches yield the same results. Additionally, a short description for the usage of the new module is provided in the Supporting Information.

2.3. Velocity Distributions. Because charge scaling usually affects the dynamical behavior more than the structural one,⁶⁵ we implemented the option to analyze the velocity of species in the pore structures into CONAN in the form of

radial velocity distributions. Similarly, as for the radial density profiles,⁴⁸ the space around the center axis of the pore is divided into distinct radial increments. The velocity of each atom within the pore is calculated and sorted into the according increments. After the entire trajectory is processed, the velocities assigned are averaged by taking the sum of all entries and dividing by the total number of entries for each grid point.

3. RESULTS AND DISCUSSION

3.1. Radial Density Analysis. **3.1.1. Mass Density.** Radial mass density profiles (Figure 2) reflect the preferential radial positions of each species and give insights into the arrangement of the IL perpendicular to the pore axis. The x -axis distances are normalized in that radial distances (d_{rad}) are divided by the radius of the CNT (r_{CNT}), and the values at the y -axis are densities ρ normalized to the reference density ρ_0 ($\rho_0(352\text{ K}) = 1.244\text{ kg m}^{-3}$).⁶⁶ We show density profiles of the individual ions together with the overall density profile (black curve in Figure 2). Further, for the $[C_2C_1Im]^+$ -cation, the density profiles of specific atoms are plotted in Figure 3.

The $[C_2C_1Im]^+$ -cation contributes most to the overall density profile because it possesses the highest molecular mass. For the smallest pore N_8 , all ions are mainly located near the center axis with three distinct maxima (see the upper left panel in Figure 2). The proximity to the center suggests that the size of the ions roughly matches the size of the CNT, leading to very limited space for perpendicular movements to the central axis. The density at the center axis is roughly twice as large as in the bulk but decays quickly and reaches 0 at $0.6\ d_{rad}/r_{CNT}$. While the cation density profile (blue curve) consists of two maxima, one at $0.3\ d_{rad}/r_{CNT}$ and one at $0.2\ d_{rad}/r_{CNT}$, the anion density profile also exhibits three maxima. One is located at the center, one at $0.3\ d_{rad}/r_{CNT}$, and one at $0.4\ d_{rad}/r_{CNT}$. Such close-lying maxima indicate that the rotation of the

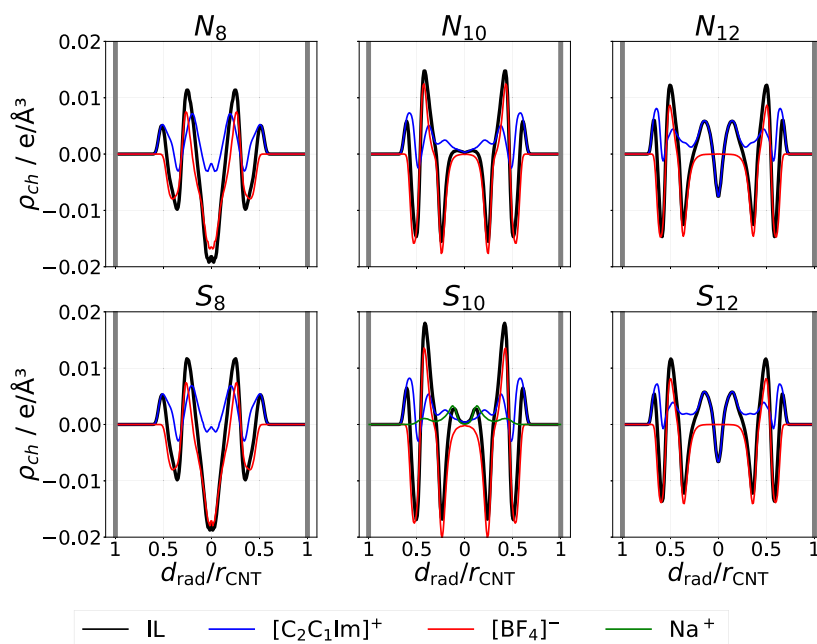


Figure 4. Radial charge density profiles of the systems N_8 – N_{12} (above) and S_8 – S_{12} (below). Black: combined charge density profile; red: $[\text{BF}_4]^-$ anions; blue: $[\text{C}_2\text{C}_1\text{Im}]^+$ cations. For system S_{10} also, a profile for the Na^+ ions is given in green. The vertical lines represent the pore wall.

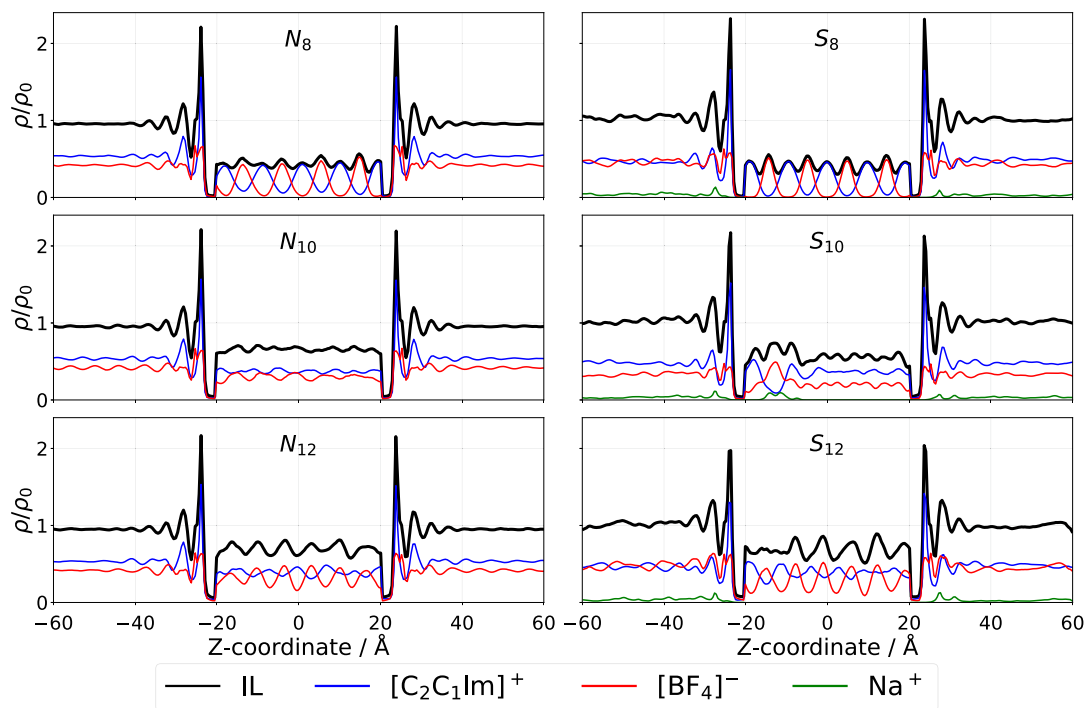


Figure 5. Axial mass density profiles of the systems N_8 – N_{12} and S_8 – S_{12} . A combined charge density profile (black curve) is shown alongside a profile for $[\text{BF}_4]^-$ (red curve) and $[\text{C}_2\text{C}_1\text{Im}]^+$ (blue curve). The centers of the pore structures are placed at 0 Å. The bulk is cut off at z distances larger than 60 Å from the center to make the comparison easier.

ions is hindered and that a specific orientation toward the walls is preferred. This is also supported by the mass density profiles of the cation, see Figure 3, where the carbon atoms in the ring and the alkyl groups show well-defined maxima. A similar behavior was found in an earlier study of the IL $[\text{C}_4\text{C}_1\text{Im}][\text{OAc}]$ in a (12,0) CNT.⁴⁸

In contrast, the ions within the pore of system N_{10} form a tubular void around the central axis. The cations occupy one

specific radial position at $0.5 d_{\text{rad}}/r_{\text{CNT}}$, while the anions can be found in two different radial positions ($0.2 d_{\text{rad}}/r_{\text{CNT}}$ and $0.5 d_{\text{rad}}/r_{\text{CNT}}$). Due to fewer local maxima, it can be assumed that both anions and cations have significantly more rotational freedom compared to those in system N_8 .

In N_{12} , two distinct layers are formed. One layer is present at the center axis and exclusively consists of cations. These cations are arranged in a chain along the center axis. The

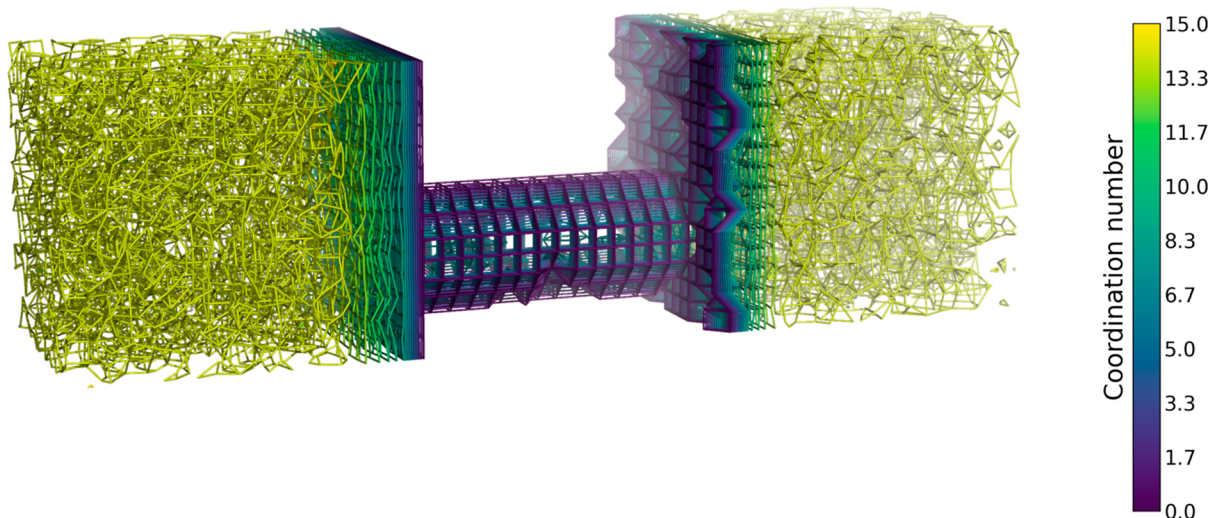


Figure 6. Three-dimensional contour plot of system N_{12} . The CN at a distance of 10 Å with the cation as reference molecule and the anion as observable is depicted in color code. For the analysis, a $15 \times 15 \times 50$ ($x \times y \times z$) grid was adjusted to the simulation box. The 3-dimensional contour plot was generated using PyMol.⁶²

second layer shows a maximum at $0.6 d_{\text{rad}}/r_{\text{CNT}}$ and forms a tubular shape. Both cations and anions contribute to the outer layer. However, the anions in this layer are slightly closer to the carbon wall with a radial density maximum at $0.6 d_{\text{rad}}/r_{\text{CNT}}$ than the cations. A second density maximum of the anions in between the outer and inner layers is found at $0.4 d_{\text{rad}}/r_{\text{CNT}}$.

When comparing the neat IL systems to those containing salt, it can be observed that the presence of the Na^+ -ions leaves the radial positions of the ions mostly unchanged. This is not the case when comparing to the charged scaled simulations; see Figure S5 in the Supporting Information. Differences to the nonscaled simulations are especially apparent for the systems containing pore 12. Both N_{12}^{sc} and S_{12}^{sc} exhibit lower maxima and higher minima compared to N_{12} and S_{12} , indicating more mobility. Interestingly, these differences are almost opposite for the smallest pore system, again indicating the constraints on the ions due to the small pore size.

3.1.2. Charge Density. Radial charge densities (ρ_{ch}) indicate the radial positions that accumulate charges inside the pore. Charge density profiles can thus give a clearer picture of the ordering of the species. While the boron atom of the anion carries a positive partial charge, the fluorine atoms are partially negatively charged. For the cation, the carbon atoms bear negative partial charges, and the hydrogen and nitrogen atoms are positively charged. Figure 4 depicts the charge density profiles for N_8 – N_{12} (above) and S_8 – S_{12} (below). As for the mass densities, we provide a combined charge density profile as well as profiles for each ion.

In contrast to the mass densities, the anions contribute more to the charge density profiles than do the cations. As only the boron atom of the $[\text{BF}_4]^-$ -anions carries a positive partial charge, its radial position is reflected in the positive values of the density profile. For all systems, the boron atom never occupies the center, but it is found at only one radial position, and this is shifted to larger $d_{\text{rad}}/r_{\text{CNT}}$ values with larger pores. Indicated by the two negative peaks in Figure 4 (red lines), the fluorine atoms are located at two distinct distances due to the geometry of the anion.

The charge density for all tube sizes is positive when the value of $d_{\text{rad}}/r_{\text{CNT}}$ is the largest. Keeping the density profile for

the $[\text{C}_2\text{C}_1\text{Im}]^+$ -cations in mind, it is clear that the position of the cations is responsible for the positive charge density. Since the nitrogen atoms are identified in radial positions below $0.5 d_{\text{rad}}/r_{\text{CNT}}$ in the radial mass density profiles, this positive contribution can be assigned to the hydrogen atoms. Interestingly, an effect due to scaling of charges on the charge density profiles is absent in all the systems, besides a decrease in the overall charge density (Figure S6 in the Supporting Information).

3.2. Axial Mass Density. Axial mass density profiles of all systems are listed in Figure 5. Again, we depict the complete density profiles and the individual ion contributions. The densities ρ are normalized to the reference density ρ_0 of the neat IL. A three-dimensional contour plot of the density distribution within the system is shown in the Supporting Information in Figure S8.

For system N_8 (upper left panel of Figure 5), the most striking feature is the alternating chain of cations and anions inside the pore. Due to the small space in this confinement, an alternating chain ordering almost like that in a solid is enforced for the ions. The larger tube sizes lack this strict ordering. In all systems, the bulk regions show the typical density oscillations near the carbon walls, which decrease in magnitude further away from the wall.^{67–69} Regarding the salt-containing systems, as mentioned before, the Na^+ ions are always located outside the tube, except for the S_{10} system. From visual inspection of the pore 8 trajectories, we observe that the Na^+ ions never penetrated the pore during the filling process. For all larger systems, the ions enter the pore during the filling process but exited them during the equilibration run. This may be explained by the different coordination behavior of Na^+ - and $[\text{C}_2\text{C}_1\text{Im}]^+$ -cations.

When comparing the systems comprising the smallest pore 8 to the larger ones, it can be seen that the observed ordering of N_8 as well of S_8 is also to some degree present in the larger pores, especially in the salt-containing systems S_{10} and S_{12} . This indicates that even if the Na^+ -ions remain outside the tube, they affect the ordering of ions inside the tube, possibly due to the increased viscosity of the IL salt mixture in bulk. An

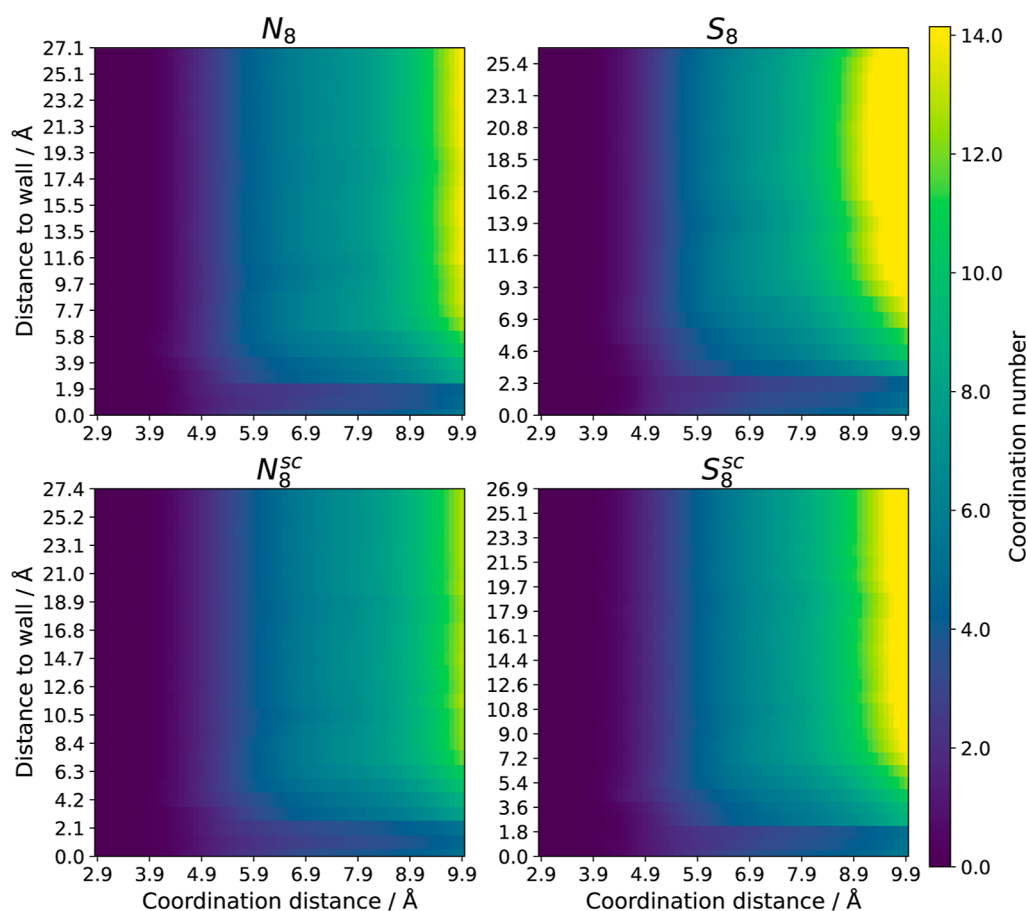


Figure 7. Heatmap of the CN at different coordination distances in relation to the distance to the wall for systems N_8 (above left), S_8 (above right), N_8^{SC} (below left), and S_8^{SC} (below right). $[C_2C_1Im]^+$ -cation: reference molecule; $[BF_4]^-$ -anion: observable.

increased viscosity upon salt addition for ILs is also observed experimentally.⁷⁰

While the IL in systems N_8 – N_{12} reach a more or less uniform density for large distances away from the carbon structures, this is to a much lesser extent the case for systems S_8 – S_{12} . In these systems, the density distribution shows small but significant oscillations within the bulk. These density fluctuations inside and outside the pore for systems S_8 – S_{12} indicate that the Na^+ ions increase the viscosity of the IL as inferred above and amplify the layering effect perpendicular to the carbon surface further into the bulk.

The same effect is visible when comparing the systems S_8 – S_{12} to S_8^{SC} – S_{12}^{SC} (Figure S7 in the Supporting Information). The aforementioned oscillating density is largely diminished for the charge-scaled systems, indicating an increased dynamics of the ions inside the pores due to reduced Coulombic attraction at reduced charges. This is consistent with the decreased dynamics for similar IL salt mixtures in the literature.^{71–74}

3.3. Coordination Number. In order to illustrate the general trends of the CN in the complete system, a 3D contour plot of the N_{12} system is exhibited in Figure 6. The CN was calculated for a maximum distance between the reference molecule and the observable of 7.72 Å. This corresponds to the distance at the first RDF minimum of the pure IL in bulk describing the first solvation shell. The shown CNs reflect exclusively the number of $[BF_4]^-$ -anions coordinating to $[C_2C_1Im]^+$ -cations. The average CN of an ion in the bulk of the liquid is around 7 as also reported in literature^{75,76} and

decreases with increasing proximity to the pore wall. The transition from bulk to wall for all systems with pore size 8 is illustrated in Figure 7. The plots for the other systems are Figures S10 and S11 in the Supporting Information.

The heatmaps show two general trends: First, systems with scaled charges have overall lower CNs (compare the upper panels with lower panels in Figure 7). Second, for coordination distances larger than the first solvation shell, systems that contain salt generally have higher CNs than the neat IL systems (compare the yellow area on the left panels with right panels in Figure 7). The first point is in agreement with the observations for the density profiles where the charge scaling results in a reduced density. The second feature is based on the fact that for distances larger than the first solvation shell, ions of the same charge are increasingly present. Na^+ ions need considerably less space than the $[C_2C_1Im]^+$ -cation, which leads to an overall higher ion density within the salt system. The space saved is partly taken up by the $[BF_4]^-$ -anions, which explains the higher CNs at distances larger than 7.72 Å.

For systems containing pore 10 (S10 in the Supporting Information) or 12 (S11 in the Supporting Information) it is also noticeable that the CN rises faster than expected for distances close to the wall from 0 to 2 Å. This is due to the small number of molecules located directly at the pore opening. For distances smaller than 2 Å, only molecules located at the pore opening are accounted for. These molecules naturally have a larger CN than molecules directly at the wall since their coordination sphere is not cut off by the

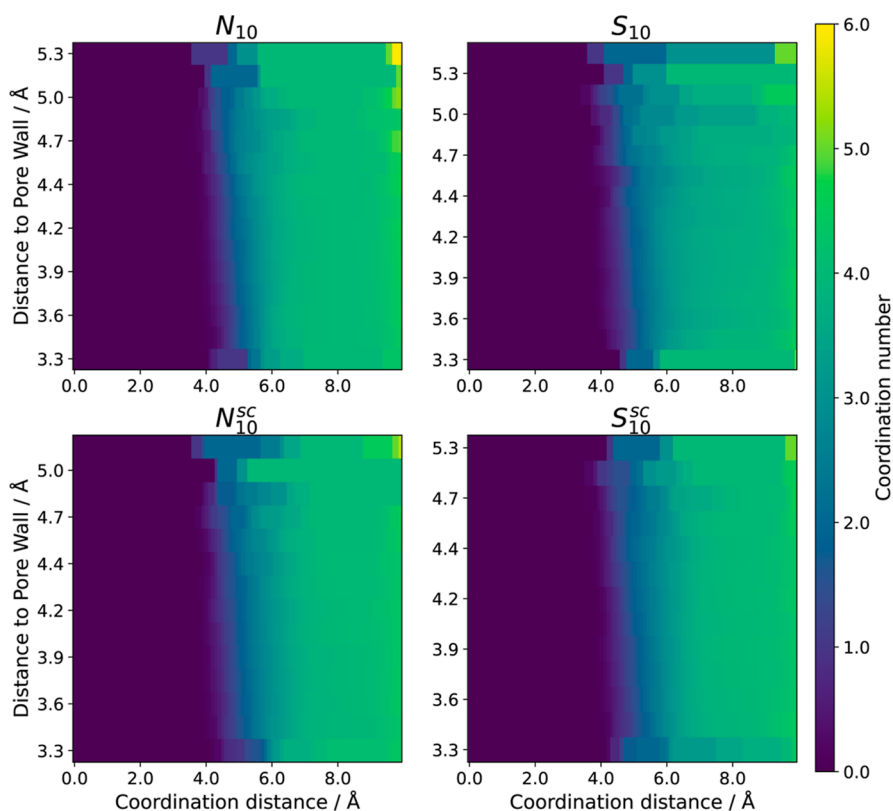


Figure 8. Cutoff of heatmap for pore 10 see Supporting Information Figure S10. $[\text{C}_2\text{C}_1\text{Im}]^+$ -cation: reference molecule; $[\text{BF}_4]^-$ anion: observable. Different coordination distances are given in relation to the distance to the pore wall for systems N_{10} , S_{10} , $\text{N}_{10}^{\text{sc}}$, and $\text{S}_{10}^{\text{sc}}$.

wall. For distances larger than 2 Å, the average CN also includes molecules that are not directly located at the pore, and the calculated CN is therefore lower. This effect is almost absent for the smallest pore since there are not many molecules in the tube and thus the CN of molecules at the pore opening and at the wall is not significantly different.

In Figure 8 a cutoff of the heatmap for pore 10, see Supporting Information Figure S10, is presented. For the systems N_{10} and $\text{N}_{10}^{\text{sc}}$, the trend that the CN is smaller for the same coordination distances when scaling the charges is apparent. This is especially present in the region between 4 and 6 Å. Such a trend is not observed when comparing the systems S_{10} and $\text{S}_{10}^{\text{sc}}$. For all systems, noise can be seen to varying degrees for the largest distances from the pore wall. This can be explained by the radial density profiles that show a density close to 0 in the center of the pore, resulting in poor sampling in these regions. In system S_{10} , this effect is particularly pronounced because the Na^+ -cations are inside the pore, which causes the $[\text{C}_2\text{C}_1\text{Im}]^+$ -cations to be slightly further away from the central axis of the pore, see Figure 3. A broader minimum for system S_{10} (compared to that of N_{10}) is apparent. When a $[\text{C}_2\text{C}_1\text{Im}]^+$ -cation is located in the center of the pore, there is a tendency toward a lower CN for the same coordination distance compared to regions closer to the wall as the anions are overwhelmingly located at larger distances near the pore wall. However, conclusions should only be drawn cautiously in this regard as the radial position of the cation is extremely rare and sampling is correspondingly poor.

3.4. Free Energy Curves. Figure 9 shows calculated free energy curves for the Na^+ -ion in the bulk and within confinement for systems S_8 – S_{12} . The region within the

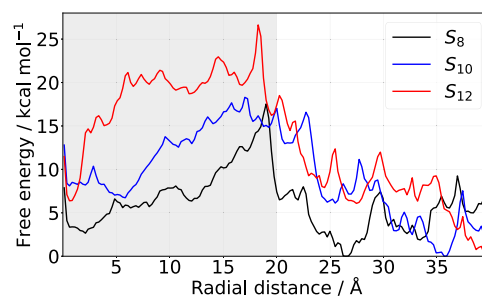


Figure 9. Free energy profiles for a Na^+ -ion in bulk and the respective pores in systems S_8 – S_{12} for radial distances of the Na^+ -ion to the center of the pore between 0 and 40 Å. The gray area describes the region within the confinement of the respective pore.

confinement is highlighted in gray. In Figure 9, the free energy is plotted against the distance of the Na^+ -ion from the center of mass of the respective pore. The comparison between this distance and the distance in the z -direction shows that the two values hardly differ (comparison in the Supporting Information). This means that the position of the Na^+ -ion varies almost exclusively in the z -direction from the starting point during umbrella sampling. The region within the bulk (at a distance >20 Å) shows that the free energy of the Na^+ -ion is subject to considerable fluctuations. These originate from the layering of the IL imposed by the surface, as discussed in Section 3.2. However, it also shows that these values should be interpreted only qualitatively and with care. A future study will provide better converged PMF curves. For all systems, a barrier of 15–25 kcal mol^{-1} must be overcome upon entering the pore. Especially for the systems S_8 and S_{12} , a strong increase of the free energy curve at the pore openings is observed. Inside

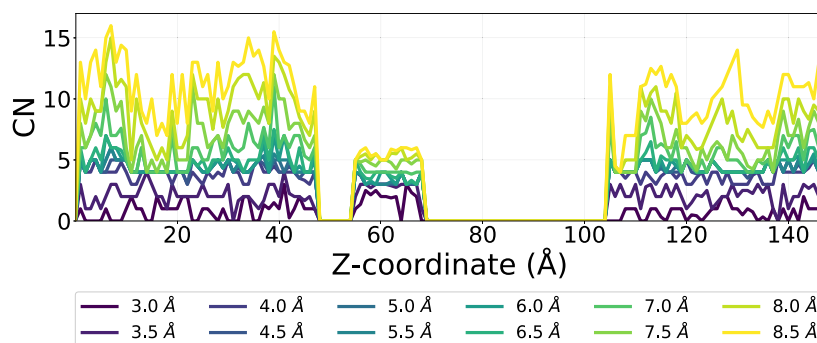


Figure 10. $\text{Na}^+ - \text{BF}_4^-$ CNs along the z -coordinate of the simulation box of system S_{10} for increasing maximal coordination distances (3.0–8.5 Å). The Na^+ ion was set as the reference species and the center of mass of the BF_4^- -anion as the observable.

the confinement, all free energy curves decrease for increasingly smaller distances, although they are on average higher than those in the bulk. This finding infers that if a Na^+ -ion were to enter the confinement, it would be trapped inside. In the previous simulations, Na^+ -ions were observed only within the confinement in system S_{10} , which is certainly due to sampling issues. For the systems S_8 and S_{12} , the energy barrier to enter the respective pores was not overcome during equilibration. However, in system S_{10} , the Na^+ -ions observed inside the pore entered the confinement during the filling process. Throughout the equilibration and production runs, the barrier at the pore opening was not overcome to leave the confinement, keeping the Na^+ -ions in a metastable state. It should be noted that in the simulations of the S_{10} system, two Na^+ -ions are present inside the pore close to each other. Due to the energetic effects of the proximity of two cations with respect to a shared coordination environment, it is conceivable that the free energy landscape would look somewhat different in this case. However, the main finding of the analysis is that, especially at the pore openings, a large energetic barrier must be overcome for the Na^+ -ion to enter and leave the confinement. As said before, these results should be taken with care, and additional simulations with better sampling must be carried out in the future.

3.5. Sodium Cation Coordination. Turning now to the Na^+ -ion coordination behavior in confinement, the CNs for different maximal coordinate distances (3.0–8.5 Å) of the Na^+ -ions as reference species and BF_4^- -anions as observable are shown in Figure 10 for system S_{10} as this was the exception with Na^+ -ions being inside the CNT.

From all plots within the figure, it is apparent that the regions along the z -coordinate of the simulation box from approximately 0 to 45 Å and 100 to 147 Å are the regions where the Na^+ cation resides in the bulk. The region from approximately 50 to 70 Å describes the Na^+ cation in the confinement of pore 10. A comparison between the progression of the graphs in the bulk area and within the confinement zone clearly shows that CN increases significantly faster within the bulk, especially for larger coordination distances. This is expected since the confined space inside the pore means that less $[\text{BF}_4]^-$ can reside in the radial vicinity of Na^+ , limited by the pore walls. However, the confined space not only causes the decrease of surrounding species but potentially also has an influence on the arrangement of ions within the first solvation shell. For this purpose, the average CN of Na^+ and $[\text{BF}_4]^-$ in the bulk (black) region and inside the pore (blue) was calculated and plotted against the coordina-

tion distance (Figure 11). For the smallest coordination distances, CN_{ave} increases faster for the region inside the pore,

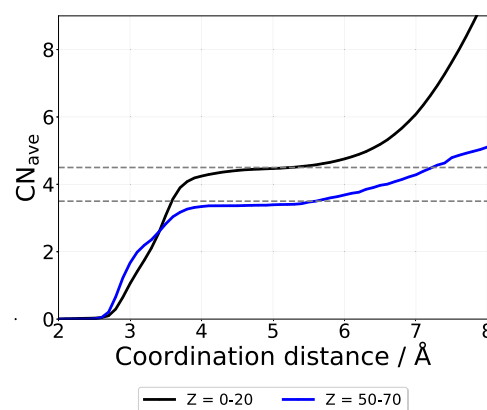


Figure 11. Average CN CN_{ave} of the Na^+ cation as reference species and BF_4^- as observable in system S_{10} . The CN was averaged in two regions. Between 0 and 20 Å (black), in the bulk of the IL–salt-mixture, and between 50 and 70 Å (blue), where the Na^+ cation is in the confinement of the pore (see Figure 10). The two horizontal lines in the plot are shown at $\text{CN}_{\text{ave}} = 3.5$ and $\text{CN}_{\text{ave}} = 4.5$.

with CN_{ave} being almost 2.0, while in the bulk CN_{ave} is about 1.0 at a distance of 3.0 Å. However, CN_{ave} increases significantly faster for increasing coordination distance within the bulk, and the plots intersect at a coordination distance of about 3.3 Å and a CN_{ave} value of 2.5. Between 3.8 and 5.2 Å, both plots reach a plateau region where CN_{ave} changes very little. This region can be considered as the outer limit of the first solvation layer. CN_{ave} thus describes the average number of $[\text{BF}_4]^-$ -anions directly coordinating to the Na^+ -ions. In the bulk the value is about 4.5, and in the confinement the value is lower at about 3.5. This provides also a possible explanation for the lack of Na^+ -ions inside the CNT.

For larger distances, above 5 Å, CN_{ave} increases again for both systems. However, the curve diverges as CN_{ave} increases significantly faster in the bulk than in the confinement. It can therefore be determined that the coordination between Na^+ and $[\text{BF}_4]^-$ changes within the confinement. While there are statistically already two $[\text{BF}_4]^-$ ions at a maximum coordination distance of 3 Å to Na^+ inside the pore, the average CN is lower in the bulk at this distance. Nevertheless, within the first solvation shell, there are on average 4.5 $[\text{BF}_4]^-$ ions in the bulk, in the confinement there are only 3.5.

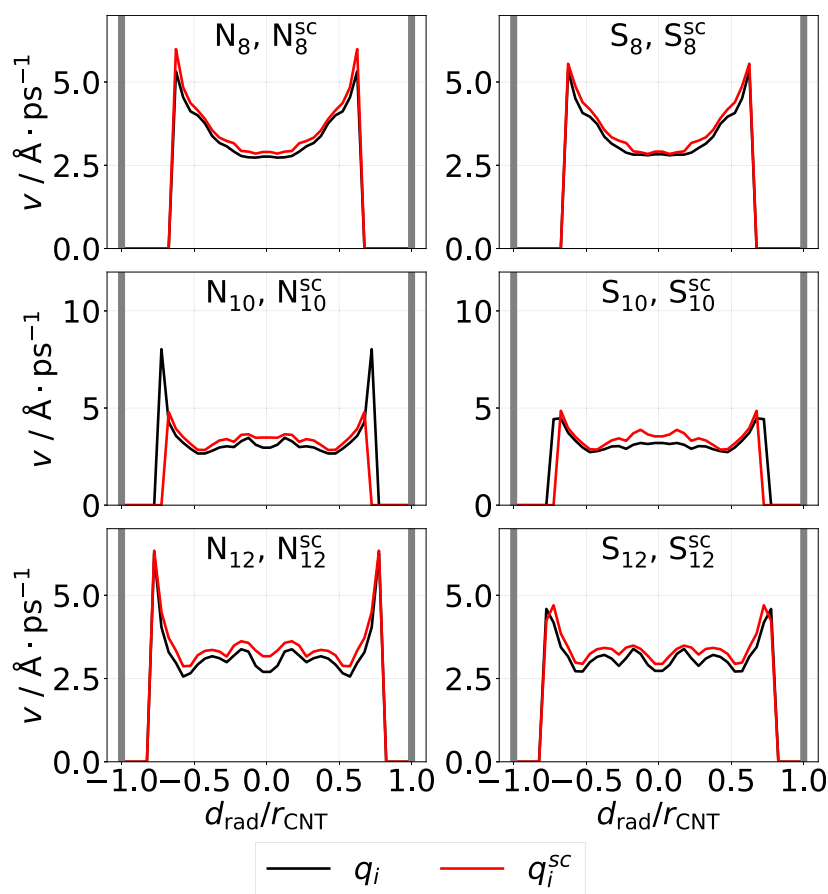


Figure 12. Radial velocity distributions of all investigated systems. The plots for the simulations with unity charges (q_i) are shown in black, with scaled charges (q_i^{sc}) in red. The velocity v is given in units of $\text{\AA} \cdot \text{ps}^{-1}$. The distances from the center axis of the pore d_{rad} are normalized using the respective pore radius r_{CNT} . The vertical bars represent the pore wall.

3.6. Radial Velocity Distributions. In Figure 12 the radial velocity distributions of all systems are depicted. Overall, the velocities increase upon scaling the force field charges (depicted in red) compared to unity charges (black). This increase is only small but present in all systems. The distribution functions for the individual pores are very similar for the neat (N) and salt (S) systems. However, it is striking that for three of the six systems examined, the maximum $d_{\text{rad}}/r_{\text{CNT}}$ values are larger for the nonscaled systems. This is surprising since the carbon atoms of the pore are not charged and no polarization effects are explicitly taken into account; the interactions between bulk and liquid in the simulations are therefore governed by van der Waals interactions. Nevertheless, there seems to be a tendency that shows that in extreme cases, the ions can move a little further away from the center axis when using unity charges, being positioned a little closer to the pore wall. Please note that in this analysis, an average velocity is determined from all the velocities calculated for each bin. In extreme cases close to the pore wall, the rarity of finding an atom at a certain radial distance from the center axis of the pore is not directly apparent from the graphs. However, it can be seen from the radial density distributions (Figure 2) that the density tends toward zero at the largest values for $d_{\text{rad}}/r_{\text{CNT}}$ of the velocity distribution function. These atom positions are therefore extremely rare.

Importantly, in addition to smaller fluctuations closer to the pore center, the velocity increases for larger $d_{\text{rad}}/r_{\text{CNT}}$ values, independent of the chosen system. This suggests that the ions

within the confinement have some rotational freedom with respect to the center axis of the pore. This is in line with the increase in velocity for larger values of $d_{\text{rad}}/r_{\text{CNT}}$ since during a rotational motion of a cluster of ions, the most displaced species have to move the fastest so that the relative distances between the species of the cluster do not change. The velocity of the individual species thus depends much more on the arrangement of the ions in the pore than on the interaction with the pore wall.

4. CONCLUSIONS

MD simulations were carried out to investigate the structural arrangement of the IL $[\text{C}_2\text{C}_1\text{Im}][\text{BF}_4]$ and its 25 mol % NaBF_4 salt mixture in confined space. The examined systems consisted of a porous carbon structure combined with a bulk phase of the studied liquids.

In most simulations, the sodium ions remained outside the confinement. Interestingly, even when the Na^+ ions were absent from the tube, the salt still had a visible effect on the ordering of the IL ions along the tube axis. While only the smallest pore size exhibited an alternating mass ordering along the tube axis in the case of the neat IL, a tubular arrangement was observed in the larger systems. In the largest pore, the ions formed two different layers, a tubular layer close to the pore wall and a chain of ions in the center. Notably, the central layer almost exclusively consisted of cations.

When focusing on the salt containing systems, the oscillating ordering of the ions was also observed for larger tube sizes.

These oscillating density profiles in the larger pore sizes vanished when scaled point charges were used, reflecting the increased dynamics inside the confinement imposed by the reduced Coulomb interactions between the ions. Furthermore, if Na^+ ions were residing inside the pore, their coordination behavior with the $[\text{BF}_4]^-$ ions was drastically changed, obviously with fewer species in the first solvation shell than in the bulk. Scaling of the force field charges increased the average velocities of the ions within the confined space. Surprisingly, in some cases, ions in nonscaled systems moved further away from the pore center. An increase in velocity toward the pore wall suggests a velocity dependence on ion arrangement rather than pore wall interactions.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.3c08493>.

Description of setup and equilibration process of the investigated systems; atomic parameters used for the analyses; partial charges of the applied force field; radial mass density and charge density profiles of systems $\text{N}_8^{\text{sc}}-\text{N}_{12}^{\text{sc}}$ and $\text{S}_8^{\text{sc}}-\text{S}_{12}^{\text{sc}}$; axial density profiles of systems $\text{N}_8^{\text{sc}}-\text{N}_{12}^{\text{sc}}$ and $\text{S}_8^{\text{sc}}-\text{S}_{12}^{\text{sc}}$; three-dimensional contour density plot of system N_{12} ; three-dimensional CN contour plot of system N_{12} and discussion on the correct application of the CN analysis; $[\text{C}_2\text{C}_1\text{Im}]^+ - [\text{BF}_4]^-$ CN heatmap of all systems containing pore 10 or 12; $\text{Na}^+ - [\text{BF}_4]^-$ CN heatmap of systems S_8-S_{12} and $\text{S}_8^{\text{sc}}-\text{S}_{12}^{\text{sc}}$; and validation of the CN analysis (PDF)

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Notes

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■ REFERENCES

- (1) Scalfi, L.; Salanne, M.; Rotenberg, B. Molecular Simulation of Electrode-Solution Interfaces. *Annu. Rev. Phys. Chem.* **2021**, *72*, 189–212.
- (2) Ge, K.; Shao, H.; Taberna, P. L.; Simon, P. Understanding Ion Charging Dynamics in Nanoporous Carbons for Electrochemical Double Layer Capacitor Applications. *ACS Energy Lett.* **2023**, *8*, 2738–2745.
- (3) Forse, A. C.; Griffin, J. M.; Merlet, C.; Carretero-Gonzalez, J.; Raji, A.-R. O.; Trease, N. M.; Grey, C. P. Direct observation of ion dynamics in supercapacitor electrodes using in situ diffusion NMR spectroscopy. *Nat. Energy* **2017**, *2*, 16216.
- (4) Forse, A. C.; Merlet, C.; Grey, C. P.; Griffin, J. M. NMR studies of adsorption and diffusion in porous carbonaceous materials. *Prog. Nucl. Magn. Reson. Spectrosc.* **2021**, *124–125*, 57–84.
- (5) Dawlaty, J. M.; Perkin, S.; Salanne, M.; Willard, A. P. The chemical physics of electrode-electrolyte interfaces. *J. Chem. Phys.* **2023**, *159*, 150401.
- (6) Bacon, C.; Serva, A.; Merlet, C.; Simon, P.; Salanne, M. On the key role of electrolyte–electrode van der Waals interactions in the simulation of ionic liquids-based supercapacitors. *Electrochim. Acta* **2023**, *455*, 142380.
- (7) Simon, P.; Gogotsi, Y. Materials for electrochemical capacitors. *Nanosci. Nanotechnol.* **2009**, *1*, 320–329.
- (8) Ma, X.; Li, C.; Martinson, A. B. F.; Voth, G. A. Water-Assisted Proton Transport in Confined Nanochannels. *J. Phys. Chem. C* **2020**, *124*, 16186–16201.
- (9) Chatzichristos, A.; Hassan, J. Current Understanding of Water Properties inside Carbon Nanotubes. *Nanomaterials* **2022**, *12*, 174.
- (10) Cicero, G.; Grossman, J. C.; Schwegler, E.; Gygi, F.; Galli, G. Water Confined in Nanotubes and between Graphene Sheets: A First Principle Study. *J. Am. Chem. Soc.* **2008**, *130*, 1871–1878.
- (11) Biriukov, D.; Kroutil, O.; Předota, M. Modeling of solid–liquid interfaces using scaled charges: rutile (110) surfaces. *Phys. Chem. Chem. Phys.* **2018**, *20*, 23954–23966.
- (12) Bouchal, R.; Al Kathemi, I.; Antonietti, M. Brønsted–Lowry Acid-Based Aqueous Eutectic Electrolyte for Practical Zinc Batteries. *Small* **2023**, *2309556*, 1–11.
- (13) Coles, S. W.; Park, C.; Nikam, R.; Kanduč, M.; Dzubiella, J.; Rotenberg, B. Correlation Length in Concentrated Electrolytes: Insights from All-Atom Molecular Dynamics Simulations. *J. Phys. Chem. B* **2020**, *124*, 1778–1786.
- (14) Borghi, F.; Piazzoni, C.; Ghidelli, M.; Milani, P.; Podestà, A. Nanoconfinement of Ionic Liquid inside Porous Carbon Electrodes. *J. Phys. Chem. C* **2021**, *125*, 1292–1303.
- (15) Park, S.; McDaniel, J. G. Interference of electrical double layers: Confinement effects on structure, dynamics, and screening of ionic liquids. *J. Chem. Phys.* **2020**, *152*, 074709.
- (16) Rajput, N. N.; Monk, J.; Singh, R.; Hung, F. R. On the influence of pore size and pore loading on structural and dynamical heterogeneities of an ionic liquid confined in a slit nanopore. *J. Phys. Chem. C* **2012**, *116*, 5169–5181.
- (17) MacFarlane, D. R.; Tachikawa, N.; Forsyth, M.; Pringle, J. M.; Howlett, P. C.; Elliott, G. D.; Davis, J. H.; Watanabe, M.; Simon, P.; Angell, C. A. Energy applications of ionic liquids. *Energy Environ. Sci.* **2014**, *7*, 232–250.
- (18) Yan, R.; Antonietti, M.; Oschatz, M. Toward the Experimental Understanding of the Energy Storage Mechanism and Ion Dynamics in Ionic Liquid Based Supercapacitors. *Adv. Energy Mater.* **2018**, *8*, 1800026.
- (19) Silvester, D. S.; Compton, R. G. Electrochemistry in Room Temperature Ionic Liquids: A Review and Some Possible Applications. *Z. Phys. Chem.* **2006**, *220*, 1247–1274.
- (20) Hessling, J.; Lange, M.; Schönhoff, M. Confinement-enhanced Li^+ ion dynamics in an ionic liquid-based electrolyte in porous material. *Phys. Chem. Chem. Phys.* **2023**, *25*, 23510–23518.
- (21) Dong, Y.; Steinhart, M.; Butt, H.-J.; Floudas, G. Conductivity of Ionic Liquids In the Bulk and during Infiltration in Nanopores. *J. Phys. Chem. B* **2023**, *127*, 6958–6968.

- (22) Lin, Z.; Goikolea, E.; Balducci, A.; Naoi, K.; Taberna, P. L.; Salanne, M.; Yushin, G.; Simon, P. Materials for supercapacitors: When Li-ion battery power is not enough. *Mater. Today* **2018**, *21*, 419–436.
- (23) Marcerou, E.; Daffos, B.; Taberna, P.-L.; Simon, P. Understanding ageing mechanism of carbon electrodes in double layer capacitors operating in organic electrolytes. *Electrochim. Acta* **2023**, *472*, 143399.
- (24) Shao, Y.; Knijff, L.; Dietrich, F. M.; Hermansson, K.; Zhang, C. Modelling Bulk Electrolytes and Electrolyte Interfaces with Atomistic Machine Learning. *Batteries Supercaps* **2021**, *4*, 585–595.
- (25) Seebeck, J.; Merlet, C.; Meißner, R. H. Elucidating Curvature-Capacitance Relationships in Carbon-Based Supercapacitors. *Phys. Rev. Lett.* **2022**, *128*, 086001.
- (26) Lahrar, E. H.; Simon, P.; Merlet, C. Carbon–carbon supercapacitors: Beyond the average pore size or how electrolyte confinement and inaccessible pores affect the capacitance. *J. Chem. Phys.* **2021**, *155*, 184703.
- (27) Costa Gomes, M.; Pison, L.; Červinka, C.; Padua, A. Porous Ionic Liquids or Liquid Metal–Organic Frameworks? *Angew. Chem., Int. Ed.* **2018**, *57*, 11909–11912.
- (28) Avila, J.; Corsini, C.; Correa, C. M.; Rosenthal, M.; Padua, A.; Costa Gomes, M. Porous Ionic Liquids Go Green. *ACS Nano* **2023**, *17*, 19508–19513.
- (29) Wang, J.; Buzolic, J. J.; Mullen, J. W.; Fitzgerald, P. A.; Aman, Z. M.; Forsyth, M.; Li, H.; Silvester, D. S.; Warr, G. G.; Atkin, R. Nanostructure of Locally Concentrated Ionic Liquids in the Bulk and at Graphite and Gold Electrodes. *ACS Nano* **2023**, *17*, 21567–21584.
- (30) Al-Masri, D.; Yunis, R.; Hollenkamp, A. F.; Pringle, J. M. A symmetrical ionic liquid/Li salt system for rapid ion transport and stable lithium electrochemistry. *Chem. Commun.* **2018**, *54*, 3660–3663.
- (31) Hakim, L.; Ishii, Y.; Matsumoto, K.; Hagiwara, R.; Ohara, K.; Umebayashi, Y.; Matubayasi, N. Transport Properties of Ionic Liquid and Sodium Salt Mixtures for Sodium-Ion Battery Electrolytes from Molecular Dynamics Simulation with a Self-Consistent Atomic Charge Determination. *J. Phys. Chem. B* **2020**, *124*, 7291–7305.
- (32) Matsumoto, K.; Hwang, J.; Kaushik, S.; Chen, C. Y.; Hagiwara, R. Advances in sodium secondary batteries utilizing ionic liquid electrolytes. *Energy Environ. Sci.* **2019**, *12*, 3247–3287.
- (33) Hagiwara, R.; Matsumoto, K.; Hwang, J.; Nohira, T. Sodium Ion Batteries using Ionic Liquids as Electrolytes. *Chem. Rec.* **2019**, *19*, 758–770.
- (34) Salanne, M.; Rotenberg, B.; Naoi, K.; Kaneko, K.; Taberna, P.-L.; Grey, C. P.; Dunn, B.; Simon, P. Efficient Storage Mechanisms for Building Better Supercapacitors. *Nat. Energy* **2016**, *1*, 16070.
- (35) Kim, H.-K.; Yu, A.; Titirici, M.; Merlet, C.; Gao, Y.; Wang, J. Materials challenges for supercapacitors. *APL Mater.* **2023**, *11*, 070401.
- (36) Salanne, M. Ionic Liquids for Supercapacitor Applications. *Top. Curr. Chem.* **2017**, *375*, 63.
- (37) Helmholtz, H. Ueber einige Gesetze der Vertheilung elektrischer Ströme in körperlichen Leitern mit Anwendung auf die thierisch-elektrischen Versuche. *Ann. Phys.* **1853**, *165*, 211–233.
- (38) Gouy, M. Sur la constitution de la charge électrique à la surface d'un électrolyte. *J. Phys. Theor. Appl.* **1910**, *9*, 457–468.
- (39) Chapman, D. L. L. A contribution to the theory of electrocapillarity. *London Edinburgh Philos. Mag. J. Sci.* **1913**, *25*, 475–481.
- (40) Stern, O. H. Zur Theorie der elektrolytischen Doppelschicht. *Z. Elektrochem. Angew. Phys. Chem.* **1924**, *30*, 508–516.
- (41) Grahame, D. C. The Electrical Double Layer and the Theory of Electrocapillarity. *Chem. Rev.* **1947**, *41*, 441–501.
- (42) Bazant, M. Z.; Storey, B. D.; Kornyshev, A. A. Double Layer in Ionic Liquids: Overscreening versus Crowding. *Phys. Rev. Lett.* **2011**, *106*, 046102.
- (43) Fedorov, M. V.; Kornyshev, A. A. Towards understanding the structure and capacitance of electrical double layer in ionic liquids. *Electrochim. Acta* **2008**, *53*, 6835–6840.
- (44) Feng, G.; Huang, J.; Sumpter, B. G.; Meunier, V.; Qiao, R. A “counter-charge layer in generalized solvents” framework for electrical double layers in neat and hybrid ionic liquid electrolytes. *Phys. Chem. Chem. Phys.* **2011**, *13*, 14723–14734.
- (45) Antonietti, M.; Chen, X.; Yan, R.; Oschatz, M. Storing electricity as chemical energy: beyond traditional electrochemistry and double-layer compression. *Energy Environ. Sci.* **2018**, *11*, 3069–3074.
- (46) Chmiola, J.; Yushin, G.; Gogotsi, Y.; Portet, C.; Simon, P.; Taberna, P. L. Anomalous Increase in Carbon Capacitance at Pore Sizes Less Than 1 Nanometer. *Science* **2006**, *313*, 1760–1763.
- (47) Merlet, C.; Péan, C.; Rotenberg, B.; Madden, P. A.; Daffos, B.; Taberna, P.-L.; Simon, P.; Salanne, M. Highly confined ions store charge more efficiently in supercapacitors. *Nat. Commun.* **2013**, *4*, 2701.
- (48) Dick, L.; Kirchner, B. CONAN - Novel Tool to Create and Analyze Liquids in Confined Space. *J. Chem. Inf. Model.* **2023**, *63*, 6706–6716.
- (49) CONAN - GitHub repository, 2023. <https://github.com/kirchners-manta/conan> (accessed December 12, 2023).
- (50) CONAN - Manual, 2023. <https://con-an.readthedocs.io/en/latest> (accessed December 12, 2023).
- (51) Plimpton, S. Fast Parallel Algorithms for Short-Range Molecular Dynamics. *J. Comput. Chem.* **1995**, *117*, 1–19.
- (52) Jorgensen, W. L.; Maxwell, D. S.; Tirado-Rives, J. Development and Testing of the OPLS All-Atom Force Field on Conformational Energetics and Properties of Organic Liquids. *J. Am. Chem. Soc.* **1996**, *118*, 11225–11236.
- (53) Jorgensen, W. L.; Tirado-Rives, J. The OPLS [optimized potentials for liquid simulations] potential functions for proteins, energy minimizations for crystals of cyclic peptides and crambin. *J. Am. Chem. Soc.* **1988**, *110*, 1657–1666.
- (54) Canongia Lopes, J. N.; Deschamps, J.; Padua, A. A. H. Modeling Ionic Liquids Using a Systematic All-Atom Force Field. *J. Phys. Chem. B* **2004**, *108*, 2038–2047.
- (55) Canongia Lopes, J. N.; Padua, A. A. H. Molecular Force Field for Ionic Liquids Composed of Triflate or Bistriflylimide Anions. *J. Phys. Chem. B* **2004**, *108*, 16893–16898.
- (56) Cole, M. W.; Klein, J. R. The interaction between noble gases and the basal plane surface of graphite. *Surf. Sci.* **1983**, *124*, 547–554.
- (57) Humphrey, W.; Dalke, A.; Schulten, K. VMD – Visual Molecular Dynamics. *J. Mol. Graphics* **1996**, *14*, 33–38.
- (58) Stone, J. E. An efficient library for parallel ray tracing and animation. M. Sc. thesis, 1747, University of Missouri–Rolla, 1998.
- (59) Leontyev, I.; Stuchebrukhov, A. Accounting for electronic polarization in non-polarizable force fields. *Phys. Chem. Chem. Phys.* **2011**, *13*, 2613–2626.
- (60) Schmollgruber, M.; Schröder, C.; Steinhäuser, O. Dielectric spectra of ionic liquids and their conversion to solvation dynamics: a detailed computational analysis of polarizable systems. *Phys. Chem. Chem. Phys.* **2014**, *16*, 10999–11009.
- (61) Grossfield, A. WHAM: the weighted histogram analysis method, version 2.0.11, 2014. http://membrane.urmc.rochester.edu/wordpress/?page_id=126 (accessed March 13, 2024).
- (62) Schrödinger, L.; DeLano, W., PyMOL, 2023. <http://www.pymol.org/pymol>.
- (63) Brehm, M.; Kirchner, B. TRAVIS - A Free Analyzer and Visualizer for Monte Carlo and Molecular Dynamics Trajectories. *J. Chem. Inf. Model.* **2011**, *51*, 2007–2023.
- (64) Brehm, M.; Weber, H.; Thomas, M.; Hollóczki, O.; Kirchner, B. Domain Analysis in Nanostructured Liquids: A Post-Molecular Dynamics Study at the Example of Ionic Liquids. *ChemPhysChem* **2015**, *16*, 3271–3277.
- (65) Frömbgen, T.; Blasius, J.; Dick, L.; Drysch, K.; Alizadeh, V.; Wylie, L.; Kirchner, B. Reducing Uncertainties in and Analysis of Ionic Liquid Trajectories. In *Comprehensive Computational Chemistry*, 1st ed.; Yáñez, M., Boyd, R. J., Eds.; Elsevier: Oxford, 2024; pp 692–722.

(66) Taguchi, R.; Machida, H.; Sato, Y.; Smith, R. L. High-Pressure Densities of 1-Alkyl-3-methylimidazolium Hexafluorophosphates and 1-Alkyl-3-methylimidazolium Tetrafluoroborates at Temperatures from (313 to 473) K and at Pressures up to 200 MPa. *J. Chem. Eng. Data* **2009**, *54*, 22–27.

(67) Wang, G. J.; Hadjiconstantinou, N. G. Molecular mechanics and structure of the fluid-solid interface in simple fluids. *Phys. Rev. Fluids* **2017**, *2*, 094201.

(68) Carter, B. M. G. D.; Royall, C. P.; Dyre, J. C.; Ingebrigtsen, T. S. Isomorphs in nanoconfined liquids. *Soft Matter* **2021**, *17*, 8662–8677.

(69) Ingebrigtsen, T. S.; Dyre, J. C. The impact range for smooth wall-liquid interactions in nanoconfined liquids. *Soft Matter* **2014**, *10*, 4324–4331.

(70) Paillard, E.; Zhou, Q.; Henderson, W. A.; Appetecchi, G. B.; Montanino, M.; Passerini, S. Electrochemical and Physicochemical Properties of PY[₁₄]FSI-Based Electrolytes with LiFSI. *J. Electrochem. Soc.* **2009**, *156*, A891.

(71) Nikitina, V. A.; Nazet, A.; Sonnleitner, T.; Buchner, R. Properties of Sodium Tetrafluoroborate Solutions in 1-Butyl-3-methylimidazolium Tetrafluoroborate Ionic Liquid. *J. Chem. Eng. Data* **2012**, *57*, 3019–3025.

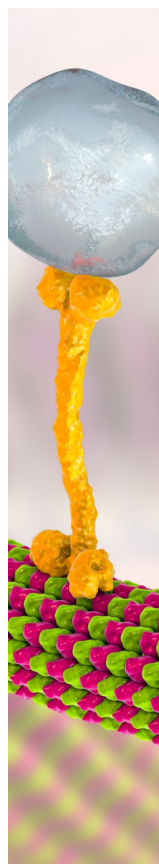
(72) Lourenço, T. C.; Zhang, Y.; Costa, L. T.; Maginn, E. J. A molecular dynamics study of lithium-containing aprotic heterocyclic ionic liquid electrolytes. *J. Chem. Phys.* **2018**, *148*, 193834.

(73) Rauber, D.; Hofmann, A.; Philippi, F.; Kay, C. W. M.; Zinkevich, T.; Hanemann, T.; Hempelmann, R. Structure-Property Relation of Trimethyl Ammonium Ionic Liquids for Battery Applications. *Appl. Sci.* **2021**, *11*, 5679.

(74) Forsyth, M.; Yoon, H.; Chen, F.; Zhu, H.; MacFarlane, D. R.; Armand, M.; Howlett, P. C. Novel Na⁺ Ion Diffusion Mechanism in Mixed Organic–Inorganic Ionic Liquid Electrolyte Leading to High Na⁺ Transference Number and Stable, High Rate Electrochemical Cycling of Sodium Cells. *J. Phys. Chem. C* **2016**, *120*, 4276–4286.

(75) Ingenmey, J.; Hollóczki, O.; Kirchner, B. Ion pairing in ionic liquids. In *Encyclopedia of Ionic Liquids*; Zhang, S., Ed.; Springer Singapore: Singapore, 2020; pp 1–14.

(76) Kirchner, B.; Malberg, F.; Firaha, D. S.; Hollóczki, O. Ion pairing in ionic liquids. *J. Phys.: Condens. Matter* **2015**, *27*, 463002.



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